

## PORTFOLIO RESEARCH

# Can Portfolio Optimization Improve the Allocation of Sucorinvest Fund Class A's Top Holdings?

Research conducted by T. Ghiyas Kausara • September 2026

## EXECUTIVE SUMMARY

Using the same ten disclosed stocks, the best portfolio found among 10,000 constrained simulations raised historical annualized return from 15.94% to 18.42%, while volatility increased only modestly from 23.30% to 23.90%. The Sharpe Ratio improved from 0.4694 to 0.5617.

| ANNUALIZED RETURN      | VOLATILITY             | SHARPE RATIO           |
|------------------------|------------------------|------------------------|
| <b>15.94% → 18.42%</b> | <b>23.30% → 23.90%</b> | <b>0.4694 → 0.5617</b> |
| +2.49 pp               | +0.60 pp               | +19.7%                 |

## 00 | RESEARCH SETUP

### OBJECTIVE

Examine whether the disclosed top-10 equity holdings could have achieved a better historical risk-adjusted return through reweighting, while keeping the stock universe unchanged.

### DATA

- Period: 1 Jan 2021 – 16 Sep 2026
- Top-10 holdings: 71.74% of NAV, normalized to 100% of the observable portfolio
- Dividend-adjusted prices; dividends assumed reinvested
- August 2026 holdings snapshot

### PORTFOLIO CONSTRAINTS

- Same 10 disclosed stocks
- Long-only, fully invested
- Maximum 13.94% per stock
- No short selling
- No transaction costs or reconstitution
- Risk-free rate: 5%

### METHOD

10,000 feasible Monte Carlo portfolios; performance evaluated using annualized return, volatility, covariance, correlation and Sharpe Ratio.

**Scope note:** This is an ex-post reweighting study of the observable top-10 holdings, not a reconstruction of the fund's full historical portfolio.

## 01 | PORTFOLIO RESULTS

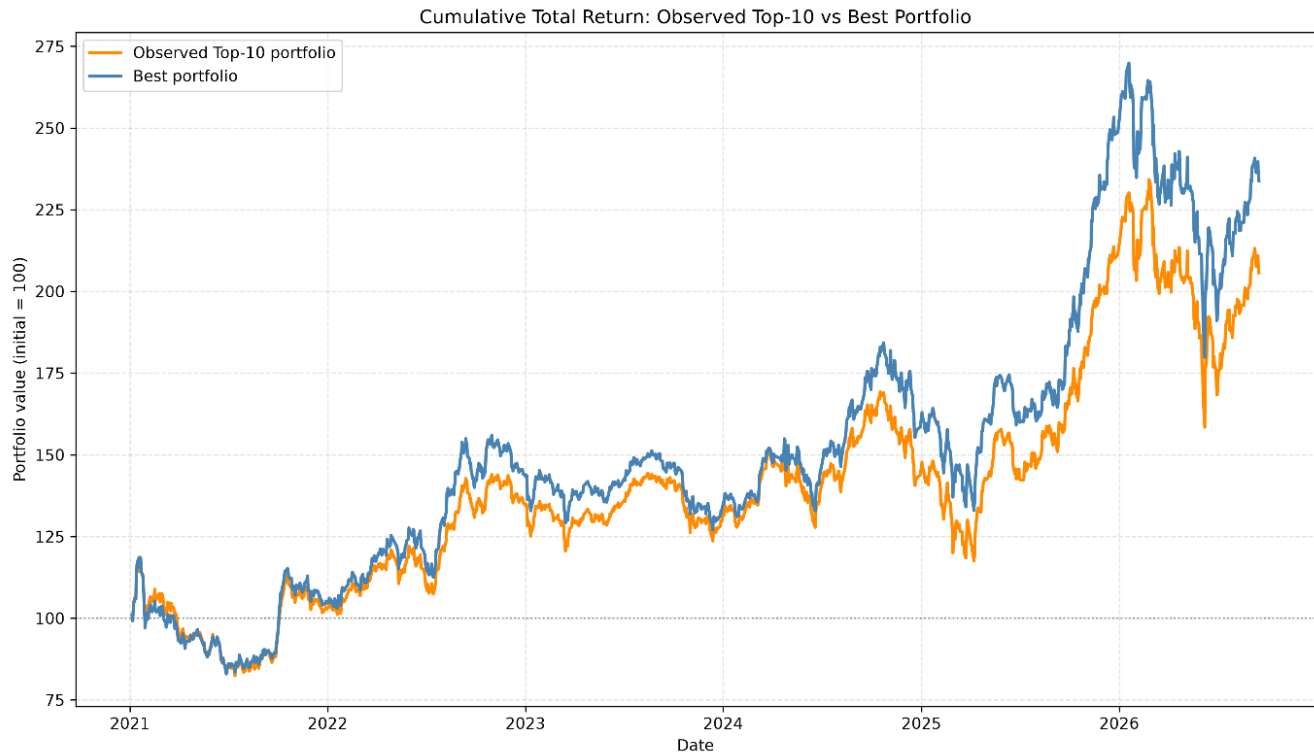


Figure 1. Ex-post cumulative growth: observed top-10 vs best simulated portfolio | Initial value = 100

| Portfolio             | Annual Return | Volatility    | Sharpe Ratio  |
|-----------------------|---------------|---------------|---------------|
| Observed Top-10       | 15.94%        | 23.30%        | 0.4694        |
| Best Simulated Sharpe | <b>18.42%</b> | <b>23.90%</b> | <b>0.5617</b> |

The best simulated portfolio improves annualized return by 2.49 percentage points while adding only 0.60 percentage points of volatility. The Sharpe Ratio rises by approximately 19.7%, indicating a more favorable historical risk-return trade-off within the same stock universe.

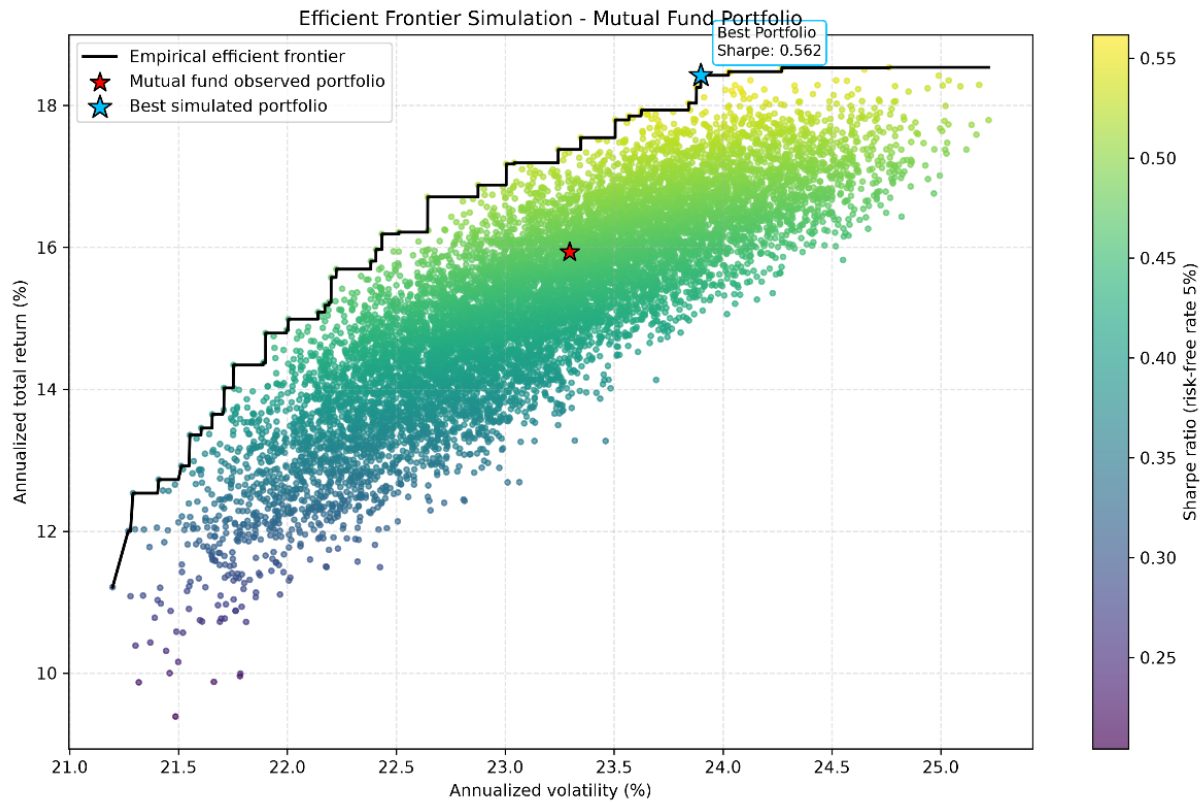


Figure 2. Efficient frontier from 10,000 feasible portfolio simulations

Simulation #4,448 produced the highest Sharpe Ratio in the 10,000-run sample: 0.5617, with an annualized return of 18.42% and volatility of 23.90%. The observed portfolio remains within the feasible opportunity set, but the simulation identifies a higher-return allocation for only a moderate increase in risk.

## 02 | DIVERSIFICATION & CORRELATION

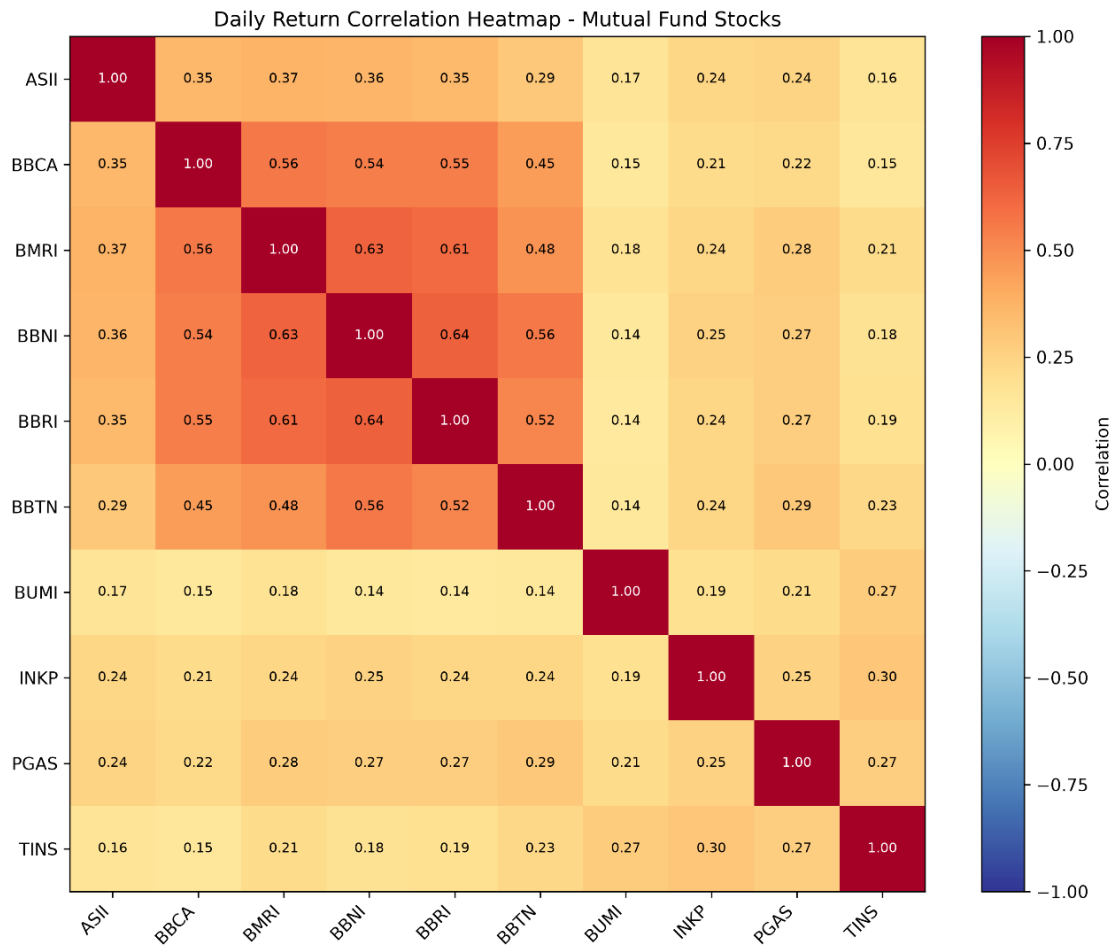


Figure 3. Daily return correlation heatmap for the disclosed top-10 holdings

| BANKING CLUSTER                                    | LOWER CO-MOVEMENT                                  | PORTFOLIO EFFECT                                                                                                           |
|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| BBNI–BBRI 0.64<br>BMRI–BBNI 0.63<br>BMRI–BBRI 0.61 | BUMI–BBRI 0.14<br>BUMI–BBNI 0.14<br>BBKA–BUMI 0.15 | All pairwise correlations remain positive;<br>diversification comes from lower co-movement<br>rather than negative hedges. |

The strongest relationships are concentrated among banking stocks, meaning that holding several banks does not provide the same diversification benefit as combining them with assets that move differently. BUMI and TINS show comparatively lower correlations with the banking group, which can help reduce portfolio co-movement.

### INTERPRETATION

Correlation is only one input. The best simulated portfolio still reduces INKP to 0.37% despite its moderate correlations, showing that allocation is driven by the combined effect of historical return, volatility and covariance—not diversification alone.

### 03 | ALLOCATION CHANGES

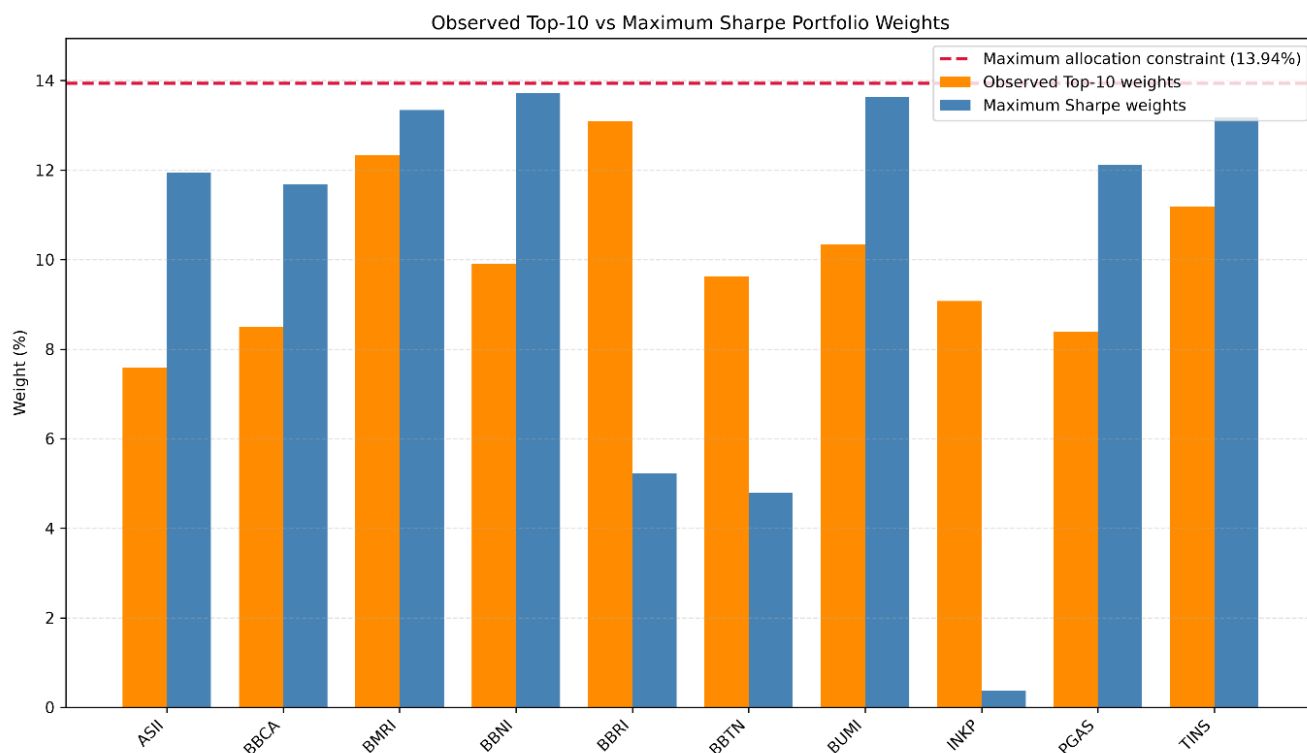


Figure 4. Observed vs best simulated portfolio weights; maximum allocation = 13.94%

| Stock | Observed Weight | Best Simulated Weight | Change |
|-------|-----------------|-----------------------|--------|
| ASII  | 7.58%           | 11.94%                | +4.36% |
| BBKA  | 8.49%           | 11.67%                | +3.18% |
| BMRI  | 12.34%          | 13.34%                | +1.00% |
| BBNI  | 9.90%           | 13.72%                | +3.82% |
| BBRI  | 13.09%          | 5.22%                 | -7.87% |
| BBTN  | 9.62%           | 4.80%                 | -4.82% |
| BUMI  | 10.34%          | 13.63%                | +3.29% |
| INKP  | 9.07%           | 0.37%                 | -8.70% |
| PGAS  | 8.39%           | 12.11%                | +3.72% |
| TINS  | 11.18%          | 13.18%                | +2.00% |

The largest reductions are INKP (-8.70 pp), BBRI (-7.87 pp) and BBTN (-4.82 pp). The largest increases are ASII (+4.36 pp), BBNI (+3.82 pp), PGAS (+3.72 pp) and BUMI (+3.29 pp). Several increased positions approach the 13.94% cap, showing that the constraint is binding for the most attractive simulated allocations.

Historical adjusted-price behavior provides context: TINS and BUMI appreciated strongly over the sample period, while INKP and BBTN ended below their adjusted starting-price levels. However, the final simulated weights also reflect volatility and cross-asset correlations, not price appreciation alone.

## 04 | KEY FINDINGS, LIMITATIONS & CONCLUSION

|                                                                                                                                                            |                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>01 RISK-ADJUSTED IMPROVEMENT</b><br>Sharpe Ratio improved from 0.4694 to 0.5617, with return rising faster than volatility.                             | <b>02 SAME STOCK UNIVERSE</b><br>The result comes from reweighting the same ten securities rather than changing stock selection.    |
| <b>03 DIVERSIFICATION MATTERS</b><br>Bank stocks are relatively highly correlated, while BUMI and TINS provide lower co-movement with the banking cluster. | <b>04 WEIGHTS MATTER</b><br>The simulation materially reduces INKP, BBRI and BBTN while increasing BUMI, BBNI, PGAS, ASII and TINS. |

### LIMITATIONS

- Only the currently disclosed top-10 holdings, representing 71.74% of NAV, are analyzed; the remaining holdings are unknown.
- August 2026 holdings are applied retrospectively to data beginning in January 2021, creating look-ahead bias and preventing interpretation as actual fund history.
- The best portfolio is the best result among 10,000 Monte Carlo simulations; it is not guaranteed to be the mathematically unique or global optimum.
- Transaction costs, taxes, liquidity, reconstitution and other fund-management constraints are excluded.

### CONCLUSION

This study shows that portfolio construction can materially influence historical risk-adjusted performance even when stock selection remains unchanged. Among 10,000 constrained simulations, the best Sharpe portfolio increased annualized return from 15.94% to 18.42% while volatility rose only modestly from 23.30% to 23.90%, improving the Sharpe Ratio from 0.4694 to 0.5617.

The result reflects a combination of stronger historical return characteristics and diversification across assets with different return behavior. Quantitative reweighting can therefore serve as a systematic complement to a fund manager's stock-selection process, but the findings remain ex-post and should not be interpreted as evidence of future outperformance.

#### BOTTOM LINE

Within the observable top-10 universe, allocation mattered: a different mix of the same stocks produced a better historical risk-return profile under the stated constraints.

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